# Lab 5 Task
# AND,OR,NOT,NOR, NAND, XOR & XNOR

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## 1. Home Security System

A homeowner wants to design a simple security system where an alarm (A) sounds when either the front door (F) or the back door (B) is open, provided that the system is armed (S).

- **Boolean variables:**
  - F (1 = front door open, 0 = front door closed)
  - B (1 = back door open, 0 = back door closed)
  - S (1 = system armed, 0 = system disarmed)
- **Question:** Write a Boolean expression for the alarm status (A) based on the door and system status, and draw the logic gate circuit for the Boolean function in proteus.
- **Solution:**

$$A = S.(F + B)$$

## 2. Water Level Control System

A water tank control system needs to fill the tank (activate the pump) if the water level is low (L) and if the pump is switched on manually (P). However, the system should not fill the tank if an overflow condition (O) is detected.

- **Boolean variables:**
  - L (1 = water level low, 0 = water level adequate)
  - P (1 = pump switch on, 0 = pump switch off)
  - O (1 = overflow detected, 0 = no overflow)
- **Question:** Write a Boolean expression for the pump control (C) based on the water level, pump switch, and overflow status, and draw the logic gate circuit for the Boolean function in proteus.

- **Solution:**

$$C = L \cdot P \cdot \overline{O}$$

## 3. Traffic Light Control

A traffic control system needs to activate a green light (G) if either the main road has traffic (M) or the side road has traffic (S), but not both simultaneously.

- **Boolean variables:**
  - M (1 = main road has traffic, 0 = no traffic)
  - S (1 = side road has traffic, 0 = no traffic)
- **Question:** Write a Boolean expression for the green light (G) based on the traffic conditions, and draw the logic gate circuit for the Boolean function in proteus.
- **Solution:**

$$G = M \oplus S$$

## 4. Elevator Control System

An elevator only operates when the door is closed (D) and either the up button (U) or the down button (N) is pressed.

**Boolean Variables:**

- D (1 = door closed, 0 = door open)
- U (1 = up button pressed, 0 = not pressed)
- N (1 = down button pressed, 0 = not pressed)

**Question**: Write a Boolean expression for the elevator operation (E) based on the door and button statuses, and draw the logic gate circuit for the Boolean function on proteus.

**Solution:**

$$E = D \cdot (U + N)$$

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## 5. Heating System Control

A heating system turns on (H) when the temperature is low (T) and the heater is enabled (E), but it should not operate if the safety override (O) is activated.

**Boolean Variables:**

- T (1 = temperature low, 0 = temperature adequate)
- E (1 = heater enabled, 0 = heater disabled)
- O (1 = safety override activated, 0 = no safety override)

**Question:** Write a Boolean expression for the heating system control (H) and draw the logic gate circuit for the Boolean function on proteus.

**Solution:** $H = T \cdot E \cdot \overline{O}$

## 6. Lighting Control System

A room's light (L) is turned on when either the main switch (M) or the motion sensor (S) is triggered, but only when it is dark (D).

**Boolean Variables:**

- M (1 = main switch on, 0 = main switch off)
- S (1 = motion sensor triggered, 0 = no motion detected)
- D (1 = dark, 0 = light outside)

**Question:** Write a Boolean expression for the lighting control (L) and draw the logic gate circuit for the Boolean function on proteus.

**Solution:** $L = D \cdot (M + S)$

## 7. Garage Door Control System

A garage door (G) opens when either the remote (R) or the button inside the garage (B) is pressed, provided the safety sensor (S) detects no obstruction.

**Boolean Variables:**

- R (1 = remote pressed, 0 = remote not pressed)
- B (1 = button pressed, 0 = button not pressed)
- S (1 = no obstruction, 0 = obstruction detected)
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Write a Boolean expression for the garage door control (G) and draw the logic gate circuit for the Boolean function and simulate it on proteus.

**Solution:**

$$G=S \cdot (R+B)$$